

EDUCATION

McGill University	Montreal, Canada
<i>B.A Computer Science, Fourth Year</i>	<i>Sep. 2019 – Present</i>
• Courses: Data Science, Software Design, Web Development, Algorithm Design • Hackathons: CTF Hackathon, HackOS-AGI Hackathon, PolyFinances-DataThon, CodeJam	

EXPERIENCE

McGill University, Computer Task Force	<u>Website</u>	Montreal, Canada
<i>Web Developer</i>		<i>Aug 2024 - Present</i>
• Developing the school’s election website using Vue.js, Node.js, HTML, CSS. • Implementing GitLab CI/CD to manage the full software development lifecycle from feature branching to automated builds, tests, deployment of the website. • Secured the voting process for 300+ students by integrating OAuth 2.0, preventing unauthorized access by enforcing McGill-only email validation. • Updated old dependencies and added automated linting/formatting to enforce project-wide coding standards.		
McGill University, Data Science Research Project	<u>GitHub</u>	Montreal, Canada
<i>Research Student</i>		<i>Jan 2025 - May 2025</i>
• Developed transformer-based bot detection system trained on AWS SageMaker, setting OpenAI’s API as a baseline, and outperforming it by 18% in accuracy (76% vs 58%). • Built a data pipeline by ingesting 100K+ tweets from Twitter API into MongoDB, cleaning textual data with Python, and loading it into PyTorch for model training.		
Korean Military		Seongnam, South Korea
<i>Intelligence Analyst</i>		<i>Sep 2021 - July 2023</i>
• Reduced data entry time by 80%, developing a Python and Selenium desktop application to automate the parsing and entry of complex geolocation data into military portals. • Designed a user-friendly GUI with Tkinter and deployed the software as an .exe with PyInstaller, enabling non-technical team members to configure automation parameters without touching the source code.		

PROJECTS

Fitness Tracker App	<u>Website</u>
• Created a searchable exercise library with 350+ exercises with muscle group, equipment, and difficulty filters for fast workout planning and a tracking dashboard to analyze history, streaks, and weekly activity. • Deployed a containerized Spring Boot/React app to AWS ECS with Fargate, using an Application Load Balancer for traffic distribution and auto-scaling to handle demand spikes. • Built a CI/CD pipeline with GitHub Actions that runs tests, builds Docker images, pushes to ECR, and deploys to ECS, with automatic rollback if anything fails. • Designed a PostgreSQL database on RDS to store workout data, optimizing queries with indexing to cut response times by 70% while keeping user, routine, and exercise data organized.	
Personal Finance Tracker	<u>GitHub</u>
• Engineered a financial dashboard in React.js, providing real-time visualizations for key metrics including categorized expenses, income, and savings rate. • Built a REST API backend (Node.js, Express, TypeScript), integrating JWT authentication with bcrypt password hashing and middleware-protected routes to ensure data privacy for users. • Implemented PostgreSQL migrations with node-pg-migrate for automated, version-controlled schema deployments.	

TECHNICALSKILLS

Coding: Java, Python, SQL, Node.js, JavaScript, TypeScript, HTML, CSS, Bash, Unix, R
Libraries: React, Swagger, Mockito, Lombok, Pydantic, Tailwind CSS, PyAutoGUI, Pytorch, Pandas, Numpy, Bokeh
Frameworks: Spring Boot, Spring Security, JUnit, JPA/Hibernate, Angular, Jest, Express, Vite, Vue.js, FastAPI
Tools: Git, Github Actions, Docker, AWS (ECS, ECR, Lambda, ALB/ELB), Selenium, MongoDB, PostgreSQL
Skills: Unit Testing, API Design, TCP/IP, SDLC, Data Wrangling, Feature Engineering